AMERICAN UNIVERSITY OF BEIRUT
MAROUN SEMAAN FACULTY OF ENGINEERING & ARCHITECTURE

Assignment: Applied Ensemble Learning

EECE 490/690 – Introduction to Machine Learning

American University of Beirut

Fall 2026–2027

Random Forests and Boosting

Submission: One fully executed .ipynb file and one short video **Total:** 100 points
Group size: Three students **Dataset:** Choose **ONE** option below

Why This Assignment

Random Forests and boosting are powerful ensemble methods, but achieving a reliable result depends on much more than fitting a model. Real datasets introduce missing or invalid values, categorical features, class imbalance, and the risk of data leakage. A good practitioner must select appropriate methods, compare them fairly, and explain the resulting predictions.

This assignment evaluates your ability to:

- Prepare realistic tabular data without leaking information from the test set.
- Explain the difference between bagging and boosting through experiments.
- Evaluate models beyond accuracy, especially when classes are imbalanced.

The emphasis is on sound decisions and clear interpretation, not on achieving the highest possible accuracy.

Approved Datasets (Choose ONE)

A. Electrical Grid Stability	Binary classification of stable versus unstable simulated grid states (about 10,000 samples). Mostly numerical features. **Kaggle dataset**.
B. Heart Disease Prediction	Binary classification using patient measurements and clinical attributes (918 samples). Includes numerical and categorical features. **Kaggle dataset**.

Required Setup (Mandatory)

Begin your notebook with a short **Experiment Setup Box**. Complete all fields before modeling. The quality of this setup is assessed in Part 1.

Experiment Setup Box
Dataset (A / B): Target column and positive class:
Dataset size: Random seed: Train/test split (suggested: 80/20):
Primary evaluation metric and reason:
Group members and individual responsibilities:

Main Tasks

Complete all seven parts in your notebook. Include code, appropriate visualizations, and a concise interpretation after each major experiment.

Part 1 – Understanding the Dataset and EDA (15 points)

1. Identify the target, feature types, dataset size, and class frequencies. Provide at least **three** informative visualizations (one must display class distribution).
2. Check for missing values, invalid entries, inconsistent data types, and features that may leak the label.
3. Establish a **DummyClassifier** majority-class baseline and explain why accuracy alone may be misleading.
4. State which metric (e.g., F1, ROC-AUC, recall) matters most for your application and explain why.

Remember: Would a classifier that always predicts the majority class be useful?

Part 2 – Preprocessing and Data Quality (20 points)

Implement two clearly documented preprocessing alternatives: **(i) a basic pipeline** using justified imputation/encoding where appropriate, and **(ii) an improved pipeline** testing a more thoughtful strategy. You are encouraged to explore techniques not covered in class for this pipeline, provided you explain them briefly in your short video. Do not remove information simply because preprocessing is inconvenient.

Dataset-specific investigation (complete only your dataset's task):

- **A – Grid:** Exclude the numerical stab score when predicting stabf: the class is derived from that score. After splitting the data, reproducibly hide 5% of values in two selected sensor features (same missingness pattern for all compared approaches). Compare simple imputation with KNN imputation. Mark these results as a *simulated missing-data experiment*.
- **B – Heart:** Investigate zero measurements in RestingBP and Cholesterol. Justify whether to retain them or treat them as unavailable, then compare an alternative using imputation and/or missingness indicators. Encode categorical attributes appropriately.

Important: Split the data *before* fitting imputers, encoders, or resamplers. Fit all preprocessing operations on training data only; inside cross-validation, refit them within each training fold.

Part 3 – Random Forests and Boosting (20 points)

Train and compare the following classifiers using identical data partitions:

1. Random Forest – bagging and randomized feature selection.
2. AdaBoost – sequentially reweights attention to difficult cases.
3. XGBoost – regularized gradient-boosted decision trees.

Keep the DummyClassifier from Part 1 as a reference. Use a modest **stratified 3-fold cross-validation** experiment to explore two hyperparameters for Random Forest (e.g., max_depth and min_samples_leaf) and two for XGBoost (e.g., max_depth and learning_rate). Limit tuning to **six candidate settings per model**. AdaBoost may use its default settings.

Guiding questions: Why can boosting outperform bagging? When can boosting overfit? Did extra complexity help?

Part 4 – Class Imbalance and SMOTE (10 points)

Investigate the positive/negative class ratio and perform **one controlled imbalance-handling experiment**:

- Compare a model trained on the original data with one using **SMOTE** for numeric data or **SMOTENC** for mixed categorical/numerical data. For mixed-type data, handle categorical features correctly rather than applying ordinary SMOTE to one-hot indicators.
- Apply resampling **only inside training folds**, using an imblearn pipeline. Never resample the final test data. Ensure any distance-dependent preprocessing is appropriate for SMOTE.
- Discuss whether resampling was actually needed. A balanced dataset may show no improvement; that is a valid finding.

Guiding question: Can recall improve while precision or accuracy worsens?

Part 5 – Fair Evaluation and Comparison (15 points)

Use validation results, not the test set, to choose a final model. Your notebook must contain:

- A comparison table showing **accuracy, precision, recall, F1, and ROC-AUC** (report the positive class and specify validation versus test results).
- A confusion matrix and one ROC or precision–recall curve for the final model.
- A short analysis of training versus validation performance and potential overfitting.
- One **final** evaluation on the untouched test set after model selection.

Guiding question: Would you select a different model if false negatives were especially costly?

Fair-comparison rule. Change one factor at a time. For example, compare Random Forest with the basic and improved pipelines *before* claiming that a performance gain comes from changing the model rather than the preprocessing.

Part 6 (BONUS) – Explainability with SHAP (+10 points)

As we discussed, tree models are known for being highly interpretable. Consequently, many ML engineers apply explainable AI (XAI) techniques on top of these models. This is a bonus section you can explore that has not been covered in class.

Apply **SHAP** to your selected tree-based classifier. Provide:

1. A **global explanation** (e.g., SHAP summary/beeswarm or mean absolute SHAP values) identifying the most influential features.
2. A **local explanation** (e.g., waterfall plot) for one individual prediction, preferably a false positive or false negative.
3. A brief discussion of whether the explanation is sensible for your application and why predictive importance does *not* establish causality.

Optional extension (up to 5 bonus points): Use LIME on the *same* individual prediction. Compare it with SHAP and discuss any disagreement; do not merely paste a second plot.

Part 7 – Conclusions and Responsible Use (10 points)

In approximately 250–400 words, answer all six questions:

1. Which model would you recommend, and why?
2. Which preprocessing decision had the greatest measured effect?
3. Did boosting improve on Random Forest? What might explain the result?
4. Did SMOTE/SMOTENC help or hurt? Why?
5. What did the model explanations reveal?
6. What is one limitation or risk that would matter in a real deployment?

Submission and Evidence Requirements

- Submit **one ZIP file** per group of three, including **one .ipynb file** with cells executed from top to bottom, readable charts, short Markdown explanations, and a fixed random seed, and **one .mp4** short video recording (5 minutes total for the team) wherein each member of the team discusses a section of the notebook and what they learned while working on it.
- Cite the Kaggle dataset and at least **two** relevant technical sources (documentation, tutorials, or papers). Note any AI-assisted code and verify its correctness.
- State member contributions. Include package requirements or versions so the notebook can be reproduced. Large dataset files need not be embedded; include download instructions.
- No lengthy report or slide deck is required. Credit is awarded for correct experimentation and interpretation, *not* for the highest accuracy.

Grading Rubric

Assessment area	Points	Full-credit expectations
Data understanding	15	Inspection, visualizations, baseline, justified metric.
Preprocessing	20	Leakage-safe pipelines; sound data-quality experiment.
Model comparison	20	Fair Random Forest/boosting comparison; bounded tuning.
Class imbalance	10	Correct SMOTE/SMOTENC application and interpretation.
Evaluation	20	Clear validation, correct plots, untouched final test.
Explainability	+10	Meaningful global/local SHAP explanations and limitations.
Conclusions	15	Evidence-based choices, limitations, and real-world risks.
Total	**100**	

Recommended references: scikit-learn documentation; imbalanced-learn documentation; XGBoost documentation; SHAP documentation.